The BLAST® Results Page

Enhanced graphical presentation with expanded functionality

https://blast.ncbi.nlm.nih.gov/

National Center for Biotechnology Information • National Library of Medicine • National Institutes of Health • Department of Health and Human Services

Introduction

BLAST search results page contains several sections and tabs. All flavors of web BLAST present similar structure and organization of the results page. Invariably, the top of the page contains headers with links, the search summary, and filtering options. The remainder of the page allows toggling between four different tabs, each serving a different purpose in examining the search results: **Descriptions**, **Graphic Summary**, **Alignments**, and **Taxonomy**. This document presents detailed annotations of each section and element of the results page, excluding the Taxonomy tab. For description of the Taxonomy tab see the following document https://go.usa.gov/xyqaY.

The Summary Section

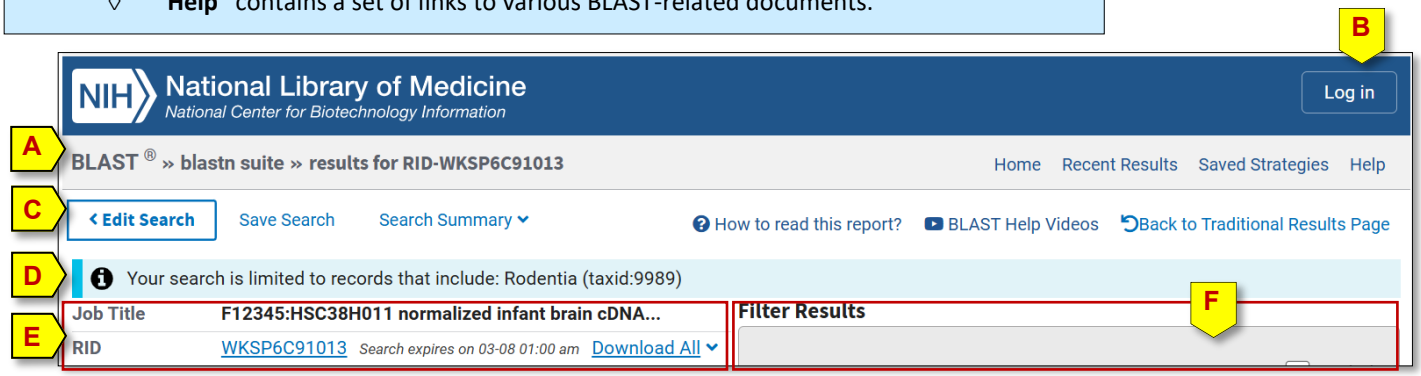
The Header

This contains three level of navigational links, the NCBI-wide set in blue at the top, the BLAST-wide set in grey is in the middle, and the BLAST result-specific set in white is at the bottom.

A. Navigational links at the top

- Breadcrumbs shows the hierarchy of the result page.
- Links to key BLAST pages:
  - ◊ "**Home**" goes back to the BLAST homepage.
  - ◊ "**Recent Results**" contains a table with a list of recently submitted BLAST searches along with key attributes and expiration date/time.
  - ◊ "**Saved Strategies**" is for saved search settings, which works best with a MyNCBI login.
  - ◊ "**Help**" contains a set of links to various BLAST-related documents.

B. Login to your MyNCBI account allows saving of settings permanently, and see all your unexpired search results done while logged in.



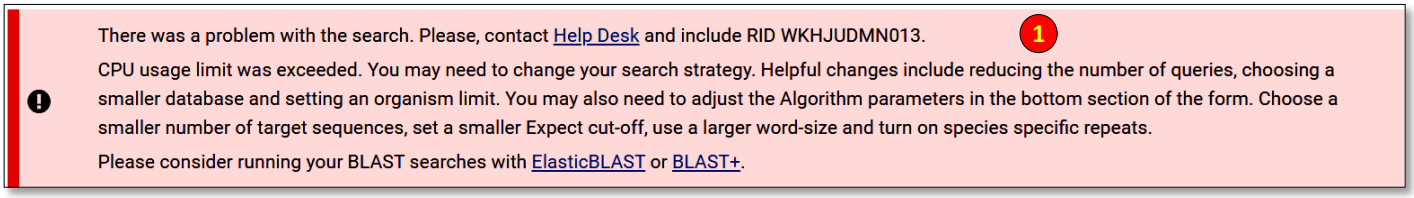
The Summary & Filter

This section has a set of links at the top (**C**). Highlighted messages or errors(**D**) appear just below it, if there is any. Below that are a set of key search attributes (**E**) at the left and a set of result filters (**F**) to the right (detailed descriptions in p2). This display of this section is always on, independent from which of the result tabs below is selected for display.

C: Top links in the Summary section

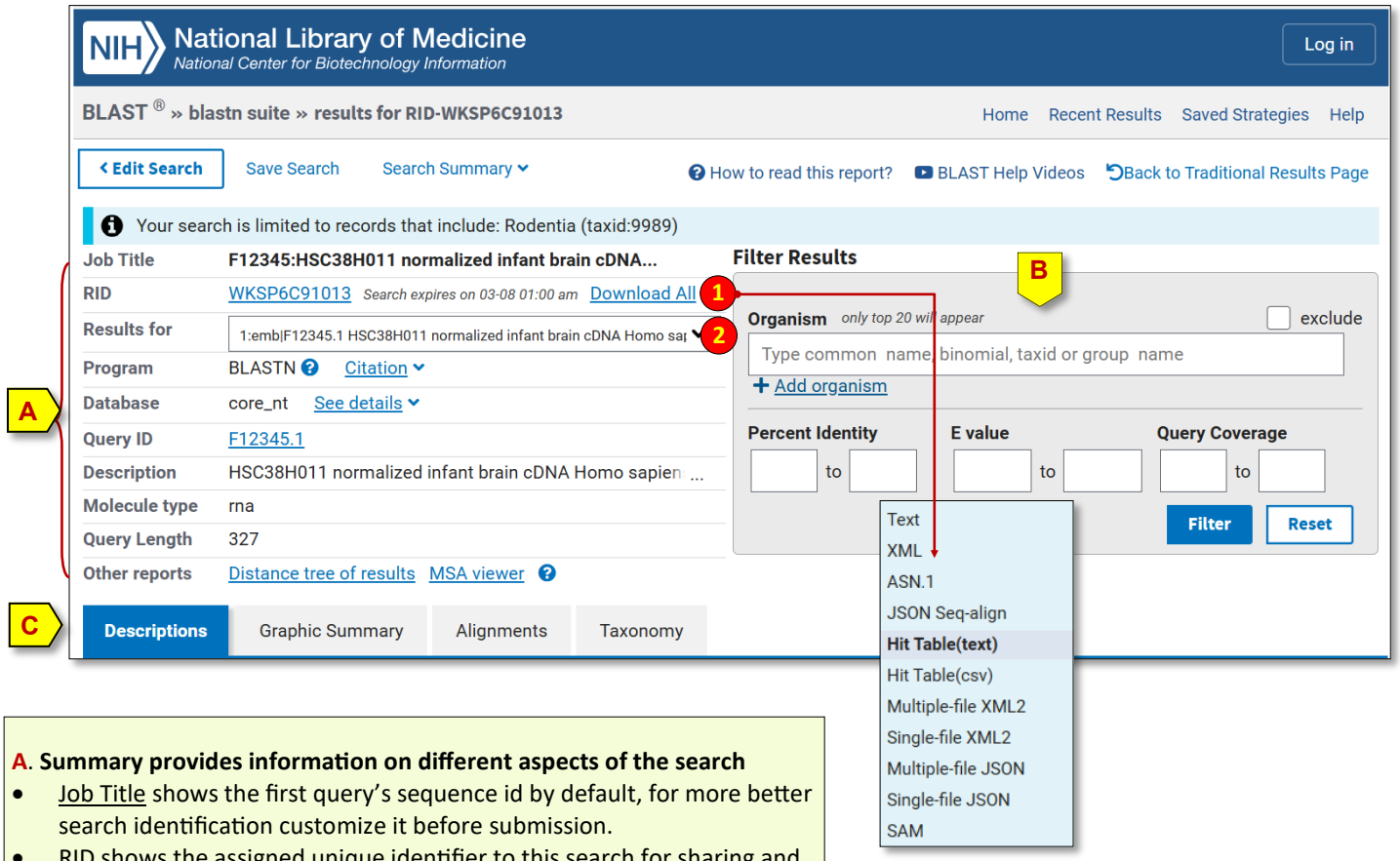
- Edit Search: brings up search page with previous settings for rerun with adjusted settings
- Save Search: saves the search settings by adding an entry to Saved Search Strategies page, do so with a MyNCBI login to save it permanently
- Search Summary: toggles open a table with summary on settings and search statistics
- How to read this report: links to this document
- BLAST Help Videos: opens a YouTube playlist of BLAST-related video in the NLM channel
- Back to Traditional Result Page: presents the result in the traditional single page format

D. If applicable, important search modifications, such as organism limit or adjustment for short query, will be displayed here. If a search failed, error messages with pink background will also be displayed here (**1**).



The Summary & Filter (cont.)

Below the link and message (described in p1 and greyed out) are the search summary on key attributes (**A**) and the result filters (**B**). This display of this section is always on, independent from which of the result tabs (**C**) further below is selected for display.



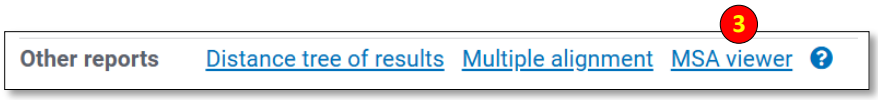
A. Summary provides information on different aspects of the search

- Job Title shows the first query's sequence id by default, for more better search identification customize it before submission.
- RID shows the assigned unique identifier to this search for sharing and problem reporting; Download All (**1**) shows a list of options to save the complete search result in desired format such as new XML (XML2), JSON, and CSV.
- Results for (**2**, not present for single query search) allows selection of a specific input query for which to show the search results.
- Program lists the search conducted, BLASTN in this case; Citation provides the reference to cite in your publication.
- Database names the target database searched; See details provides a text description of the database content.
- Query ID is the query's sequence id for which the result is shown. For query in accession format, it is linked to the record in NCBI database.
- Description is the title of the query sequence from its FASTA defline.
- Molecule type shows the query's sequence type.
- Query length is the length of the current query.
- Other reports lists links to other report formats not integrated to the new tab-based display. Protein BLAST results will have an additional "Multiple alignment" link (**3**), which sends the query and matches to multiple alignment search with COBALT, an NCBI multiple alignment tool.

B. Filter Results

- Organism allows input of organism name (select from auto-suggested list) to filter the returned hits. Check the exclude checkbox to exclude selected organism from display. Click Add organism to add additional input field.
- Percent Identity allows filtering of hits by degree of identity, such as between 94.74 and 94.76 percent.
- E Value filters result by hits' significance in Expect value, such as between 0.0001 and 5e-120 (scientific notation of $5x10^{-120}$).

Enter the desired values, then click Filter to update the result display. The Filter works through post-processing and does not change the search result identified by the assigned RID. Click Reset to see the initial set of results.



The Descriptions Table

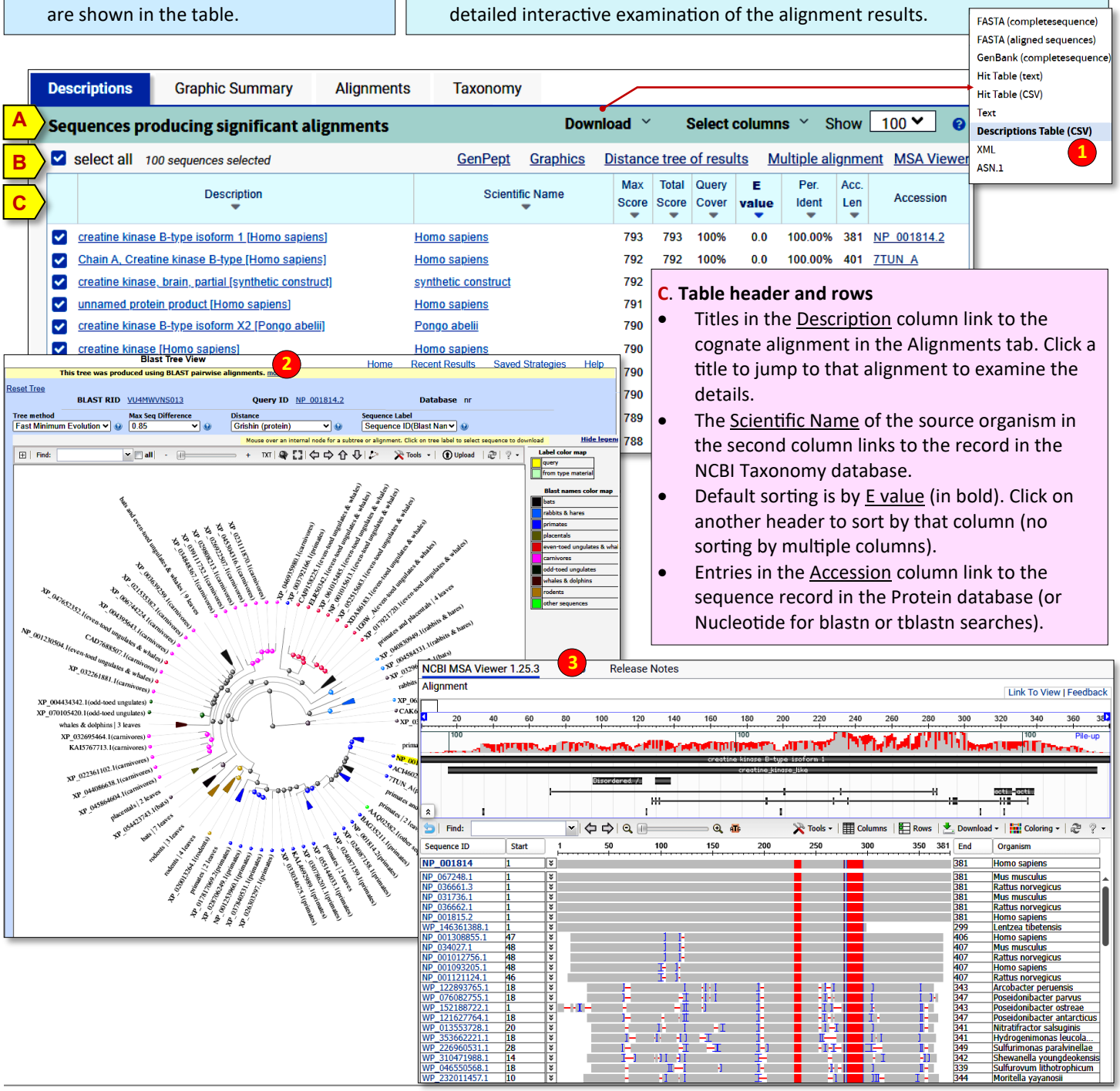
This is the default tab and contains a summary table of hits found by BLAST. Just above the summary table are three sets of pull-down menus (**A**) and a set of links to present matches found (**B**) in newly spawned browser tab. The table header (**C**) describes the column content and allows sorting by selected column.

A. Pull-down menus at the top

- Download offers options to download checkbox-selected hits in various format, includes this table itself (**1**). This option is not available from Download All, p2 (**1**).
- Manage Columns lists all available columns of the table so you can customized the columns displayed.
- Show allows you to select how many hits are shown in the table.

B. Functional links for this search

- Select All toggle on and off checkboxes for all the hits. Click on individual checkboxes to toggle selection of specific entries.
- GenPept (GenBank for nucleotide) retrieves the full records for selected match from the Protein database.
- Graphics links to a sequence viewer-based graphical view of matches anchored by the query.
- Distance tree of results presents the matches in a distance tree format (**2**).
- Multiple alignment sends query and matches for multiple alignment in Cobalt.
- MSA Viewer opens the multiple sequence alignment viewer (**3**) and allows detailed interactive examination of the alignment results.



C. Table header and rows

- Titles in the Description column link to the cognate alignment in the Alignments tab. Click a title to jump to that alignment to examine the details.
- The Scientific Name of the source organism in the second column links to the record in the NCBI Taxonomy database.
- Default sorting is by E value (in bold). Click on another header to sort by that column (no sorting by multiple columns).
- Entries in the Accession column link to the sequence record in the Protein database (or Nucleotide for blastn or tblastn searches).

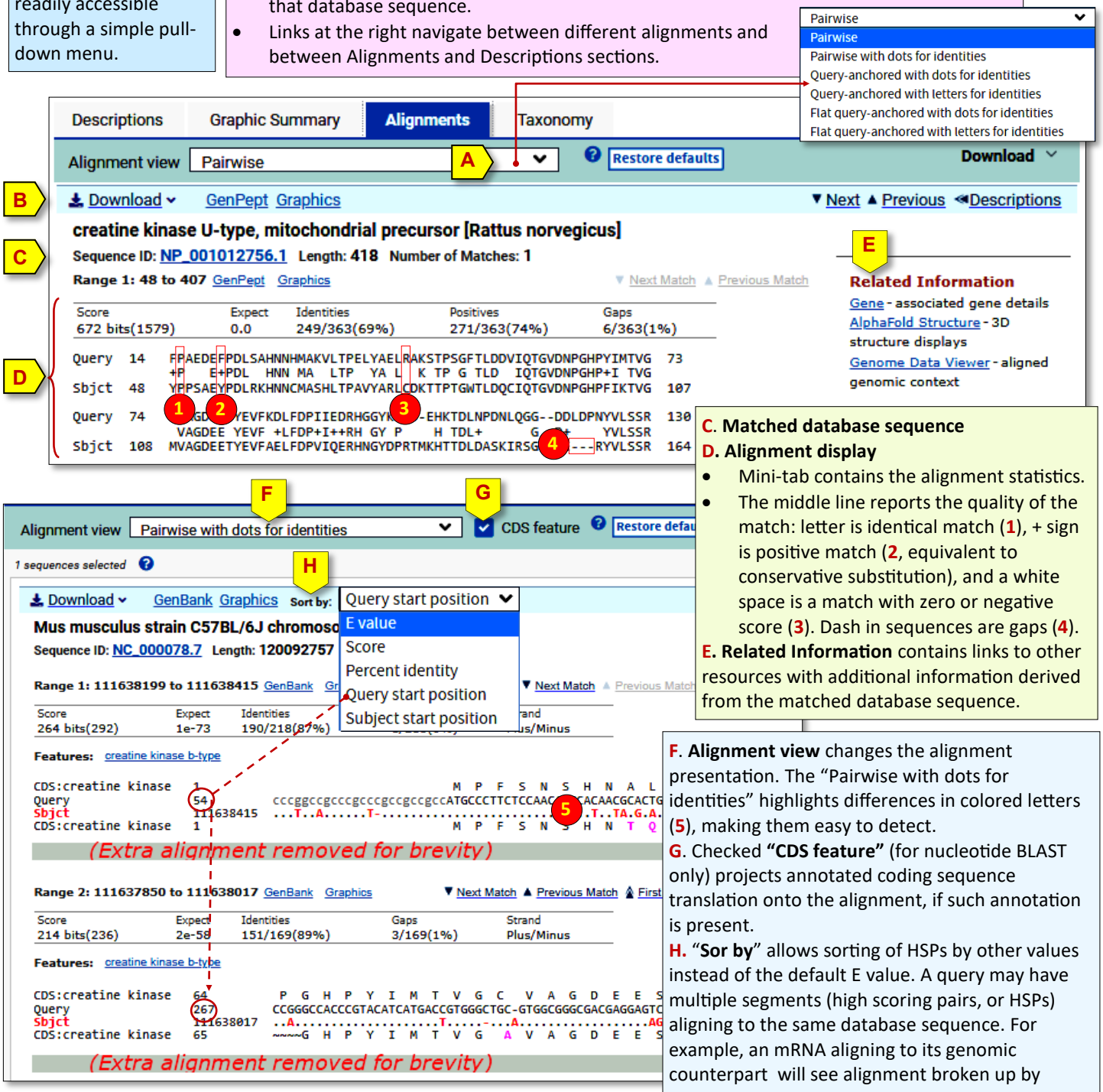
The Alignments Tab

The Alignments section (below) contains the detailed pairwise alignments between query and its matched sequences.

A. Alignment view
It makes alternative displays of the alignment results readily accessible through a simple pull-down menu.

B. Separator for matched database sequences

- Download lists formats available to get this matched database sequence.
- GenePept (**GenBank** for nucleotide) retrieves the record from the NCBI Protein database.
- Graphics links to a sequence viewer-based graphical display of the alignment, anchored by that database sequence.
- Links at the right navigate between different alignments and between Alignments and Descriptions sections.



C. Matched database sequence
D. Alignment display

- Mini-tab contains the alignment statistics.
- The middle line reports the quality of the match: letter is identical match (**1**), + sign is positive match (**2**, equivalent to conservative substitution), and a white space is a match with zero or negative score (**3**). Dash in sequences are gaps (**4**).

E. Related Information contains links to other resources with additional information derived from the matched database sequence.

F. Alignment view changes the alignment presentation. The "Pairwise with dots for identities" highlights differences in colored letters (**5**), making them easy to detect.

G. Checked **"CDS feature"** (for nucleotide BLAST only) projects annotated coding sequence translation onto the alignment, if such annotation is present.

H. "**Sor by**" allows sorting of HSPs by other values instead of the default E value. A query may have multiple segments (high scoring pairs, or HSPs) aligning to the same database sequence. For example, an mRNA aligning to its genomic counterpart will see alignment broken up by introns, leading to multiple HSPs. Re-sorting the display by "Query start position" facilitates the interrogation of alignments by their natural/ biological order on the mRNA.

Notes

For Taxonomy Tab, see this existing handout https://go.usa.gov/xyqaY.
We love to hear from our users. Please send your comments and feedback to: blast-help@ncbi.nlm.nih.gov.